

Spring Tool Suite 4

```
1 package com.example.controller;
2
3 import com.example.entity.Product;
4
5 @RestController
6 @RequestMapping("/api/products")
7 public class ProductController {
8
9     private final ProductService productService;
10
11     // Constructor-based Dependency Injection
12     public ProductController(ProductService productService) {
13         this.productService = productService;
14     }
15
16     @GetMapping
17     public List<Product> getAllProducts() {
18         return productService.getAllProducts();
19     }
20
21     @GetMapping("/{id}")
22     public ResponseEntity<Product> getProductById(@PathVariable Long id) {
23         return ResponseEntity.ok(productService.getProductById(id));
24     }
25
26     @PostMapping
27     public ResponseEntity<Product> createProduct(@RequestBody Product product) {
28         return ResponseEntity.ok(productService.saveProduct(product));
29     }
30
31     @PutMapping("/{id}")
32     public ResponseEntity<Product> updateProduct(@PathVariable Long id, @RequestBody Product productDetails) {
33         return ResponseEntity.ok(productService.updateProduct(id, productDetails));
34     }
35
36     @DeleteMapping("/{id}")
37     public ResponseEntity<Void> deleteProduct(@PathVariable Long id) {
38         productService.deleteProduct(id);
39         return ResponseEntity.noContent().build();
40     }
41 }
42
```

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```
1 package com.example.entity;
2 import jakarta.persistence.*;
3
4 @Entity
5 @Table(name = "products")
6 public class Product {
7     @Id
8     @GeneratedValue(strategy = GenerationType.IDENTITY)
9     private Long id;
10     private String name;
11     private String description;
12     private double price;
13     private int quantity;
14
15     // Constructors
16     public Product() {}
17     public Product(String name, String description, double price, int quantity) {
18         this.name = name;
19         this.description = description;
20         this.price = price;
21         this.quantity = quantity;
22     }
23
24     // Getters and Setters
25     public Long getId() { return id; }
26     public void setId(Long id) { this.id = id; }
27     public String getName() { return name; }
28     public void setName(String name) { this.name = name; }
29     public String getDescription() { return description; }
30     public void setDescription(String description) { this.description = description; }
31     public double getPrice() { return price; }
32     public void setPrice(double price) { this.price = price; }
33     public int getQuantity() { return quantity; }
34     public void setQuantity(int quantity) { this.quantity = quantity; }
35 }
36
```

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